

Printing Shapes

Topics: nested for-loops, patterns, and shapes

Lab 4: https://maryash.github.io/135/labs/lab_04.html

Nested for-loop

In order to print a basic rectangular shape, we can use a nested for-loop. You may have seen this already in cs127. Here's how to make a 5x5 rectangular shape:

```
# include <iostream>
using namespace std;
int main() {
    for(int i = 0; i < 5; i++){ // for-loop for rows
        for(int j = 0; j < 5; j++){ // for-loop for columns
            cout << "*"; // print stars (5 times total)
        }
        cout << endl; // move to next row after printing 5 stars
    }
    return 0;
}
```

Nested for-loop (continued)

The outer loop variable(*i*) corresponds to the current row and the inner loop variable(*j*) corresponds to the current column. If you are trying to create a particular shape, look for patterns in the numbers (*i,j*). For a basic rectangular shape, we print a star in every combination of *i* and *j*:

<i>i, j</i>	<i>j</i> = 0	<i>j</i> = 1	<i>j</i> = 2	<i>j</i> = 3	<i>j</i> = 4	Output
<i>i</i> = 0	0,0	0,1	0,2	0,3	0,4	*****
<i>i</i> = 1	1,0	1,1	1,2	1,3	1,4	*****
<i>i</i> = 2	2,0	2,1	2,2	2,3	2,4	*****
<i>i</i> = 3	3,0	3,1	3,2	3,3	3,4	*****
<i>i</i> = 4	4,0	4,1	4,2	4,3	4,4	*****

Checkerboard Shape

Let's try to create a checkerboard shape within a 5x5 square. Based on the table to the right, observe the relationship between the i and j values.

When do we print a star and when do we print a space?

i, j	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$	Output
$i = 0$	0,0	0,1	0,2	0,3	0,4	*****
$i = 1$	1,0	1,1	1,2	1,3	1,4	*****
$i = 2$	2,0	2,1	2,2	2,3	2,4	*****
$i = 3$	3,0	3,1	3,2	3,3	3,4	*****
$i = 4$	4,0	4,1	4,2	4,3	4,4	*****

Checkerboard Shape (continued)

Stars are printed for the following cells:

0,0	0,2	0,4	1,1	1,3	2,0	2,2	2,4	3,1	3,3	4,0	...
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Spaces are printed in the following cells

0,1	0,3	1,0	1,2	1,4	2,1	2,3	3,0	3,2	3,4	4,1	...
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Pattern: Stars are printed when both i and j share the same parity (odd/even). Spaces are printed when they have the opposite parity.

We can check the parity with the modulo operator($\%$). If i and j are both even, then $i\%2 = 0$ and $j\%2 = 0$, if they are both odd, $i\%2 = 1$ and $j\%2 = 1$. Based on this, we print a star “*” if $(i\%2 == j\%2)$, else we print a space “ ”.

Checkerboard Shape (another way)

What if we added i and j ?

Stars are printed at these cells

i, j	0,0	0,2	0,4	1,1	1,3	2,0	2,2	2,4	3,1	3,3	4,0	...
$i + j$	0	2	4	2	4	2	4	6	4	6	4	...

Spaces are printed in the following cells

i, j	0,1	0,3	1,0	1,2	1,4	2,1	2,3	3,0	3,2	3,4	4,1	...
$i + j$	1	3	1	3	5	3	5	3	5	7	5	...

Cross Shape

Similar to the checkerboard problem, can you see a pattern for the cross shape? Observe the i and j when stars are being printed.

Hint: There are two different patterns you need to find.

i, j	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$	Output
$i = 0$	0,0	0,1	0,2	0,3	0,4	*****
$i = 1$	1,0	1,1	1,2	1,3	1,4	* *
$i = 2$	2,0	2,1	2,2	2,3	2,4	*
$i = 3$	3,0	3,1	3,2	3,3	3,4	* *
$i = 4$	4,0	4,1	4,2	4,3	4,4	*****

Cross Shape (continued)

Stars are printed for the following cells:

0,0	0,4	1,1	1,3	2,2	3,1	3,3	4,0	4,4	...
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Pattern 1: Observe the diagonal going to the bottom right, its combinations are (0,0), (1,1), (2,2), and so forth. This means a star is printed when i is equal to j .

Pattern 2: Similar to the checkerboard, let's add the remaining pairs:

i, j	0, 4	1, 3	2,2	3,1	4,0
$i + j$	$0 + 4 = 4$	$1 + 3 = 4$	$2 + 2 = 4$	$3 + 1 = 4$	$4 + 0 = 4$

The column size is 5, which makes the last index of columns equal to 4.

Thus combining the patterns, we print a star if $(i == j \text{ or } (i+j) == \text{col_size}-1)$

Cross Shape (another way)

Stars are printed for the following cells:

0,0	0,4	1,1	1,3	2,2	3,1	3,3	4,0	4,4	...
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Pattern 2: Observe the combinations (0,4), (1,3), (2,2), (3,1), (4,0). We know that the outer for loop increments i by one which is why i goes from 0 to 4 naturally. However we can note that j is decreasing by 1 each iteration. We also know the column size is 5. The last index of j is one less than the column size. By subtracting i from each column size-1, we can get the new j :

i, j	0, 4	1, 3	2, 2	3, 1	4, 0
$\text{col_size} - 1 - i = j$	$5 - 1 - 0 = 4$	$5 - 1 - 1 = 3$	$5 - 1 - 2 = 2$	$5 - 1 - 3 = 1$	$5 - 1 - 4 = 0$

Thus, combining the patterns, we print a star if $(i == j \text{ or } (\text{col_size} - 1 - i) == j)$

Upside down trapezoid(ish) shape

The following is a 4x7 upside down trapezoid shape: (according to Task F)

i, j	j = 0	j = 1	j = 2	j = 3	j = 4	j = 5	j = 6	Output
i = 0	0,0	0,1	0,2	0,3	0,4	0,5	0,6	*****
i = 1	1,0	1,1	1,2	1,3	1,4	1,5	1,6	*****
i = 2	2,0	2,1	2,2	2,3	2,4	2,5	2,6	***
i = 3	3,0	3,1	3,2	3,3	3,4	3,5	3,6	*

Upside down impossible shape

Certain shapes are impossible to make, such as a 5x7 trapezoid

i, j	j = 0	j = 1	j = 2	j = 3	j = 4	j = 5	j = 6	Output
i = 0	0,0	0,1	0,2	0,3	0,4	0,5	0,6	*****
i = 1	1,0	1,1	1,2	1,3	1,4	1,5	1,6	*****
i = 2	2,0	2,1	2,2	2,3	2,4	2,5	2,6	***
i = 3	3,0	3,1	3,2	3,3	3,4	3,5	3,6	*
i = 4	4,0	4,1	4,2	4,3	4,4	4,5	4,6	???

Upside down trapezoid shape (cont)

In each row, we have to print some spaces followed by some stars. For the 4x7 shape, notice how the number of stars and spaces change on each row:

row (i)	# of stars	# of spaces before/after
0	7	0
1	5	1
2	3	2
3	1	3

Thus, we can have two for loops for each row. One to print spaces, and another to print stars. The number of spaces and stars can be tracked using int variables